



Ethical Issues in Memory Modification Technology: A Scoping Review

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Abstract Memory modification technology (MMT) refers to the use of neurotechnologies to intervene in memories. Many scholars have reflected on the ethical issues in MMT, but a comprehensive review of this topic has not been seen. This article presents the first scoping review study of ethical issues in MMT using a bibliometric and systematic approach. After thorough examination, 133 records of key literature are included in this scoping review. Six core ethical themes are extracted: (1) self, identity, and authenticity; (2) autonomy and informed consent; (3) welfare and happiness; (4) safety and risks; (5) responsibility and obligation; (6) social and legal justice. More detailed analyses are conducted on the moral stances and reasons held by different scholars concerning these ethical themes. As can be seen, current debates exhibit certain shortcomings, including ambiguous ethical concepts and a restricted scope of analysis. Therefore, we call for deeper reflections on the philosophical foundations, more precise definitions of ethical terms, and more comprehensive examinations of neurotechnological applications, in order to better address the ethical challenges in the future.

Keywords Memory · Modification · Technology · Neuroethics · Scoping review

Background

Borrowing a phrase from Hermann Ebbinghaus (1908), we can say that memory has “a long past but only a short history.” Philosophers have studied the metaphysical natures and epistemological functions of memory since antiquity.¹ However, it was not until the late nineteenth and early twentieth centuries that the emergence of psychology pushed memory studies into the scientific paradigm. For example, Atkinson and Shiffrin (1968) proposed the ESR model of memory, which states that there are three different phases of memory (encoding, storage, and retrieval) and distinguishes between different kinds of memory such as short-term memory and long-term memory. In addition, many scientists have examined the phenomena of forgetting, memory biases, false memory, and pathological amnesia (Wixted 2004; Coles and Heimberg 2002; Brainerd and Reyna 2005; Ryan et al. 2000). These scientific theoretical and empirical works have fueled philosophical discussions about

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¹ A very detailed overview of the philosophical history of memory from Plato and Aristotle to Heidegger and Ricoeur can be found in Part IX of “History of the philosophy of memory” in *The Routledge Handbook of Philosophy of Memory*. See Bernecker and Michaelian (2017).

memory, increasing the interdisciplinarity of this field (Michaelian and Sutton 2017). On the metaphysical side, philosophers are concerned with the relationship between memory, truth, time, causation, and similar concepts (Bernecker 2008; Hoerl and McCormack 2001). On the epistemological side, philosophers are dealing with the justification problem of memory contents and debating about the internalism or externalism of memory beliefs (Michaelian 2016; Senor 2019; Frise 2023).

For moral philosophers, memory is also a fascinating subject, which involves issues such as the duty of remembering and the right of forgetting. As neuroscience and technology continue to advance, the ethical problem of memory modification has gradually become a significant issue of concern for bioethics and neuroethics. In fact, it can be said that memory modification has “a long past” as well, with literary plots found in Homer’s *Odyssey*, Cicero’s *De Oratore*, and Shakespeare’s *Macbeth* (Muriithi 2014; Alonso Fernández 2019; Tan and Lim 2020). Modern philosophical works often take the form of thought experiments, such as British ethicist Derek Parfit’s discussion of “Venice’s memory” in *Reasons and Persons*. The story goes like this. One day, Jane awoke in a hospital recovery room after a successful memory operation. She had requested to replicate Paul’s memories of his Venice vacation in her own brain. Now, lying in bed, a completely new, vivid, and crystal-clear set of memories unfold in her mind:

She seems to remember walking on the marble paving of a square, hearing the flapping of flying pigeons and the cries of gulls, and seeing light sparkling on green water. One apparent memory is very clear. She seems to remember looking across the water to an island, where a white Palladian church stood out brilliantly against a dark thundercloud. (Parfit 1984)

After recounting this story, Parfit raises a question: “What should Jane believe about these apparent memories?” Parfit’s constructed scenario seems to suggest that intervening in memory raises complex and profound moral problems, such as the authenticity of self-experience and the impact of memory alterations on personal identity.

Some readers might view such thought experiment as resembling a science fiction tale or nothing

more than a conceptual game.² However, at present, the landscape of memory modification is gradually becoming a reality. For instance, in 2012, the research team led by Nobel laureate Susumu Tonegawa used optogenetics to control memory engram cells and implanted false fear memories in mice (Liu et al. 2012). In clinical applications, the U.S. Food and Drug Administration (FDA) has approved two categories of drugs, namely cholinesterase (ACHE) inhibitors and N-methyl-D-aspartate (NMDA) receptor antagonists, to enhance the memory abilities of Alzheimer’s disease patients (Dekkers and Rikkert 2007; Cummings et al. 2014). Additionally, in the treatment of patients with post-traumatic stress disorder (PTSD), some doctors prescribe medications, such as propranolol, for off-label use, to weaken traumatic memories in patients (Henry et al. 2007; Bell 2008; Liivoja and Kroes 2020). An increasing number of neuroscientists are now utilizing optogenetic technology to study the mechanisms and patterns of learning and memory and exploring more possibilities for modifying memories of model organisms (Guskjolen et al. 2018; Keene and Masek 2012; Layfield et al. 2023; Perusini et al. 2017).

From this, it can be seen that modifying memories is no longer just a fanciful idea. In fact, we have already developed a range of neurotechnologies for memory modification, including optogenetic technology, neuropharmaceuticals (propranolol, zeta inhibitory peptide, cortisol, etc.), brain stimulation technology (deep brain stimulation, transcranial magnetic stimulation, transcranial direct current stimulation, etc.), neurosurgical implants, and more. The use of neurotechnology for intervening in memory is referred to as memory modification technology (MMT).³

² A classic film about memory modification is *Eternal Sunshine of the Spotless Mind* (2004), starring Kate Winslet and Jim Carrey, in which the main characters undergo memory modification by a memory erasure company. This film can be seen as a philosophical thought experiment, and many philosophical articles have been written discussing it (Grau 2006; Levy 2007, 2011; Miah 2009; Paiella 2020; Tillman 2016).

³ In some articles, “memory modification technology” is also alternatively stated as “memory manipulation technology,” “memory modulation technology,” or “memory modification techniques,” with the abbreviation MMT used for all of these terms. However, in terms of frequency, “modification” is more commonly used, and its initial ethical implications seem more neutral compared to “manipulation.” See Erler 2011; Farina

Table 1 Different kinds of MMTs

Objectives	Forms	Meaning
Memory capacity	enhancing (Muriithi 2014) blocking (Burbiel 2015)	Improving future memory capacity Inhibiting future memory capacity or preventing memory forming
Memory content	altering (Kolber 2014), editing (Lavazza 2019) enhancing (Alonso Fernandez 2020), improving (Vedder 2010) adding (Leuenberger 2022), implanting (Zawadzki & Adamczyk 2021a) attenuating (Henry et al. 2007), blunting (Erler 2011), dampening (Kolber 2008), disassociate (DePergola II 2018a) reducing (Liao & Sandberg 2008), suppressing (Wasserman 2004), weakening (Liao 2017) deletion (Hayse & Roskies 2021), erasure (Levy 2011), eradication (Kroes & Liivoja 2019), forget (Kolber 2006), silencing (Tan & Lim 2020), wiping (Grau 2006)	Changing memory content Making past memories clearer Adding false segments of memory content Making past memory content and its emotional impact more blurred Removing segments of past memory completely

Moreover, MMT can be further categorized based on the specific objectives of application: some impact individual future capacity of memory, while others affect the content of existing memories (see Table 1).

With the development of MMT, related ethical issues are also growing. For example, a commentary in the *Financial Times* points out that the development of MMT raises concerns about the possibility of manipulating memories to achieve certain purposes: “some saw the utilitarian merit in it, while others felt uncomfortable with the deception required” (Ahuja 2016). Additionally, *New York Times* assessed that MMT holds promise for individuals with deficits to take charge of their own lives, but “there are angels and demons buried there, in addition to facts and names” (Carey 2018).

Those comments from the media have demonstrated our core concern about MMT: for the sake of well-being, we should support the use of MMT, but this might bring about safety concerns for users and societal risks. On the other hand, if we consider these potential threats and risks, we should oppose the use

of MMT, but for those with cognitive impairments or those plagued by painful memories, this means they will lose a significant opportunity for pursuing their better lives.

Faced with such a moral dilemma, based on a recognition of this technology as a double-edged sword (Gerlai and Rose 2003; Boals and Schuettler 2011), numerous scientists and ethicists have engaged in extensive academic discussions within the interdisciplinary framework of neuroethics, an academic field that inherits the principles of traditional bioethics while also providing reflections on the responsible application of cutting-edge neuroscience and technologies (Glannon 2019; Northoff 2009). Utilizing neuroethics as paradigm, various scholars have used different moral reasoning to arrive at varying viewpoints or ethical stances that should be adopted regarding MMT.

Although some review articles in this area has been published at present, it can be seen that past research has not used systematic and bibliometric research methods and has tended to focus on a single or a small number of topics rather than portraying the full range of MMT ethical issues (Sandberg and Liao 2008; Hui and Fisher 2015; Liao 2017; Tan and Lim 2020; Zawadzki 2022). This study aims to fill this gap, not only depicting the current academic

Footnote 3 (continued)
and Lavazza 2022; Hui and Fisher 2015; Leuenberger 2022; Liao and Sandberg 2008; Zawadzki 2022.

landscape of MMT ethics, sorting out different moral stances and reasons we currently have, but also providing resources for ethicists to engage in further reflection and initiate the next stage of discussion.

Method

Research Framework

The article adopts the method of scoping review to examine the relevant literature on the ethical issues in memory modification technology, sorting out the research scope, depth, and perspectives. Ethical reflections on MMT often take place under the disciplinary umbrella of neuroethics; the relevant research paradigms of this cross-discipline can be categorized into the neuroscience of ethics and the ethics of neuroscience (Roskies 2002; 2021). With regard to methodologies, neuroethics employs two distinct research approaches: empirical and normative research (Northoff 2009; McDougall 2015). In the former approach, researchers deal with real-life ethical problems encountered in the actual clinical use and scientific research of MMT by way of case studies and try to provide solutions and policy responses (Vedder and Klaming 2010; Newman et al. 2010; Burbiel 2015; Adamczyk and Zawadzki 2020). In the latter approach, researchers tend to focus on moral philosophical analyses of the ethical consequences of MMT and reflect on the moral acceptability of MMT from different normative concepts, theories, and principles (Lavazza 2019; Vukov 2022; Leuenberger 2022).

In normative research, due to differences in researchers' positions, values, and argumentative approaches, reviewers find it challenging to provide a consistent evaluation of the literature from a single objective perspective. Therefore, systematic reviews of quality face difficulties in handling such literature in this field. Scoping reviews address this challenge by including as much relevant literature on the topic as possible, anchoring the research progress in a particular subject within the field of neuroethics and the varied reasoning and value orientations of different scholars (Parsons and Johal 2022). Therefore, a scoping review is an appropriate method for examining the research progress on the ethical issues of emerging neurotechnologies. It helps us assess the scope of

ethical issues currently discussed by scholars and the quantity of available literature. Additionally, it aids in clarifying researchers' ethical stances and reasons regarding MMT (Munn et al. 2018).

To be specific, this paper will use the scoping review research framework proposed by Hilary Arksey and Lisa O'Malley (2005). They propose that scoping reviews involve the following five stages:

Stage 1: identifying the research question

Stage 2: identifying relevant studies

Stage 3: study selection

Stage 4: charting the data

Stage 5: collating, summarizing, and reporting the results

Arksey and O'Malley emphasize that these five stages of research are not linear but rather an iterative process involving reflection. Researchers can repeat the steps to ensure comprehensive coverage of the relevant research literature.

Search Strategy

This scoping review aims to address the following question: *What ethical issues are of concern in memory modification technology?* Guided by this question, this study was conducted from October to November 2023, with subsequent revisions from March to May 2024. Literature was collected from five reference databases, including four academic literature databases (Web of Science, Scopus, MEDLINE, Philosopher's Index) and one grey literature database (Google Scholar). The timeframe for literature retrieval was not set with a specific starting point, as memory modification is a relatively new neurotechnology. The endpoint was set as the retrieval date (October 30, 2023). Before the author submitted the revised manuscript based on the comments from the journal's reviewers (March 2024), the author conducted another search, checked, and supplemented relevant literature in the same search strategy.

During the retrieval phase, advanced searches were conducted in the databases using Boolean logic connectors and truncation symbols. The search strategy employed in the Web of Science database was "(TI=(*memory title words*) AND (*modification technology title words*)) AND (TS=(*ethics topic words*))." In other words, retrieval terms related to

memory and modification technology appear in the title, with ethical terms as the topic, to exclude irrelevant search records (see Table 2).

This search strategy was translated into other formats to match the syntax of Scopus, MEDLINE, and Philosopher’s Index databases for retrieval. For practical reasons, search in Google Scholar was conducted to examine literature with titles containing either “memory modification” or “memory manipulation,” two most representative keywords in this field, to supplement grey literature in this field. Additionally, consultation with experts in information management and scholars in bioethics and the philosophy of cognitive science was undertaken. Their input was sought for search strategy advice and recommendations for relevant literature to avoid overlooking important documents. Additionally, during the revision process, recommendations from Litmaps, a literature retrieval software, were also obtained. The literature encompasses handbooks, reports, books, and papers. All results obtained from these search strategies were imported into Zotero 6.0.30 software for the next phase of filtering.

Inclusion and Exclusion Criteria

In Zotero 6.0.30 software, articles identified through the aforementioned search strategies underwent two rounds of screening to assess their relevance to the research topic. The first screening focused on titles and abstracts, while the second screening involved a comprehensive evaluation of the full text.

Inclusion criteria is as follows: (i) discussing memory modification technology; (ii) addressing relevant ethical issues; (iii) utilizing philosophical speculation or based on scientific research or experiential

case studies in clinical application; (iv) published in English.

Exclusion criteria is as follows: (i) not exploring memory modification technology— that is, the core focus was not on neurotechnology but rather on methods such as psychological suggestion, hypnosis, mnemonic, or other approaches; (ii) not involving discussions of ethical issues; (iii) merely mentioning the terms “ethics” or “moral” without providing a more in-depth analysis; (iv) not published in English.

It should be noted that this study includes reviews and responses to target articles, as these commentary reviews also contribute the authors’ perspectives and present supportive or opposing views on the target articles.

Analytical Method

Furthermore, the selected literature was further organized, and an analysis was conducted on bibliometrics, ethical themes, and moral positions. The bibliometric description conforms to the requirements of Arksey and O’Malley (2005) for the chart presentation in the fourth step of a scoping review, providing an intuitive display of the quantity and connections of relevant studies. The ethical theme description aims to present the ethical issues in MMT that scientists and ethicists are concerned about. It directly responds to the core question of this study, clarifying the significant ethical issues in MMT. The analysis of moral positions aims to present the distribution of scientists’ and ethicists’ moral attitudes toward MMT. In other words, after anchoring relevant ethical issues, this article examines the judgements and reasons for the moral acceptability of MMT by different scholars. This helps us comprehend the progress in research in

Table 2 Search strategy in Web of Science database

Memory title keywords	memory OR memories OR remember* OR forget* OR *trauma* OR “PTSD”
AND	
Modification and technology title keywords	manipulat* OR modul* OR modif* OR moderat* OR enhanc* OR suppress* OR edit OR dampen* OR blunt* OR delet* OR erasure OR eradicat* OR interven* OR optogenetic* OR neurotechnolog* OR propranolol OR drug* OR alter* OR therap*
AND	
Ethics topics keywords	*ethic* OR moral* OR authenticity OR identity OR autonomy

this field and the related moral reasoning and provides some reference for our future moral judgements.

Results

Bibliometric Analysis

According to the retrieval strategy, a total of 2584 records were generated from the databases, with fifty-three records recommended by the reviewers of this manuscript and a literature review software, making a total of 2637 records. Among them, eight hundred duplicate publications were removed, along with 249 non-research articles, 110 articles not written in English, and twelve articles without title information or having been retracted, resulting in a total of 1466 documents for review. In the initial screening, 1290 irrelevant documents were excluded based on titles and abstracts. Subsequently, a full-text reading and

second screening were conducted on the remaining 176 records. Among these, forty-three articles did not meet the inclusion criteria. The literature screening process is shown in Fig. 1. In the end, 133 records are included in this scoping review. These records include sixty-seven journal articles (50.4 per cent), fifty-three review and commentary papers (39.8 per cent), twelve books and book chapters (9.0 per cent), and one dissertation (0.8 per cent). The list of key records can be seen in Supplementary Information.

The earliest academic discussions on the ethical issues of MMT originated from 2002, and a report published in 2003 from the (U.S.) President's Council on Bioethics became the targets of subsequent debates (Rose 2002; Wolpe 2002; Gerlai and Rose 2003; President's Council on Bioethics 2003; Farah et al. 2004) (see Fig. 2). This topic has since sparked broader discussions over the past decade, with the level of interest generally showing an upward trend (see Fig. 3). Peaks in the number of publications

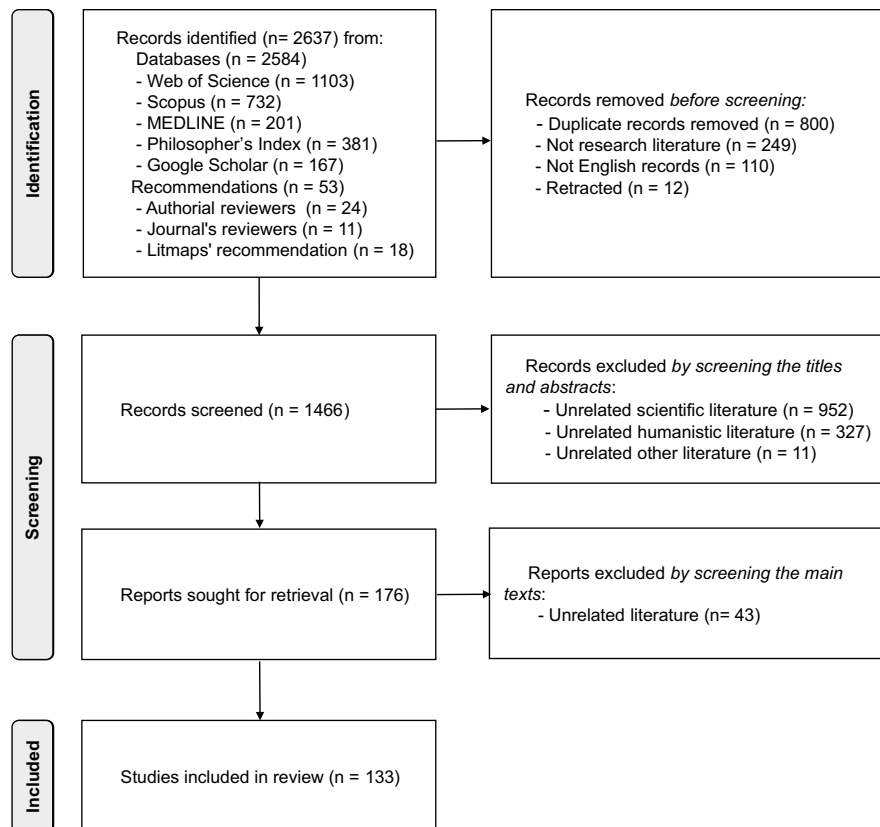


Fig. 1 PRISMA flow diagram search results

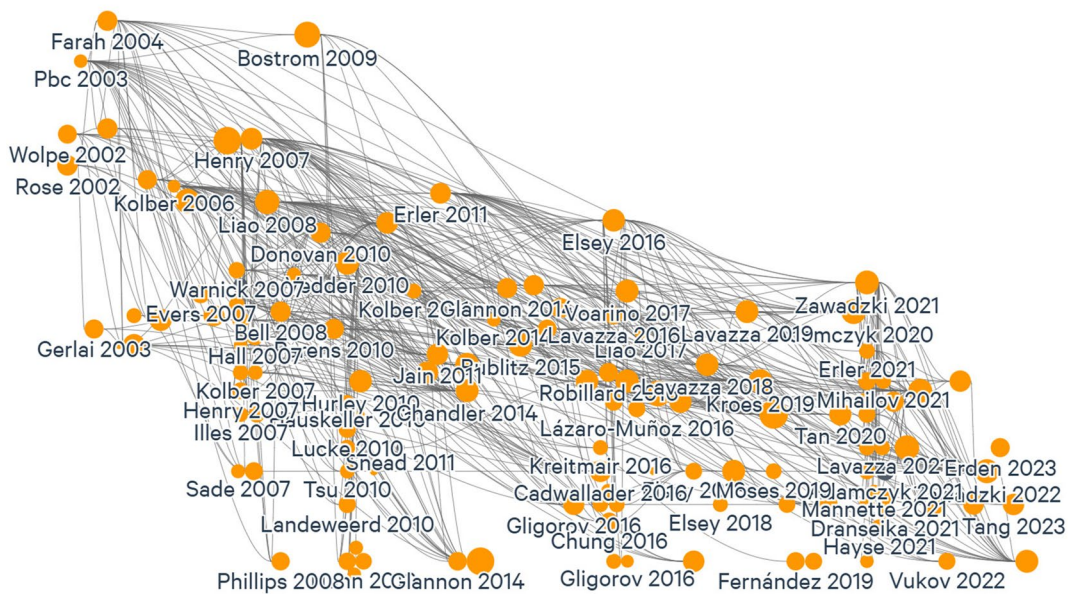


Fig. 2 Literature citation network in the field of MMT ethics

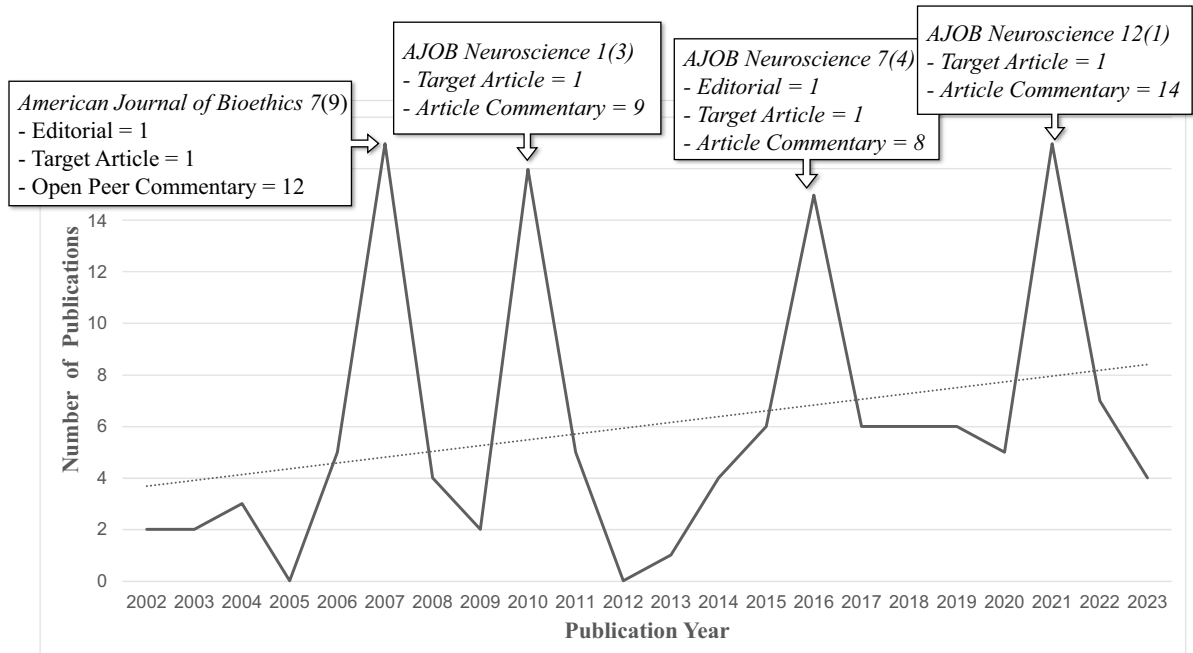


Fig. 3 Annual trend chart of publications in the field of MMT ethics

occurred in 2007, 2010, 2016, and 2021. These are attributed to the issues discussing the ethical problems related to MMT published in September 2007 in the *American Journal of Bioethics* and in June 2010, April 2016, and December 2021 in the *AJOB Neuroscience*.

Ethical Themes

The selected works address a diverse range of core ethical topics. Using VOSviewer 1.6.18 software, the author created a co-occurrence map of keywords from the aforementioned 133 records. The size of nodes represents the frequency of keyword occurrences, and the connections between nodes indicate the associations between keywords. It is evident that

topics such as identity, authenticity, and autonomy are highly emphasized issues in this field (see Fig. 4).

Moreover, all the files of records included in the study are imported into NVivo 20 software for automatic coding processing by sentence. NVivo automatically coded three themes: “memory,” “moral,” and “ethical.” Under the “memory” theme are a total of 2280 subthemes, such as “traumatic memories,” “emotional memories,” and “painful memories.” Under the “moral” theme are 506 subthemes, including “moral responsibility” and “moral judgement.” Under the “ethical” theme are 484 subthemes, including “ethical issues” and “ethical concerns.” Table 3 below shows the top ten subthemes and their reference quantities for each theme. Furthermore, by using NVivo 20 software to automatically code emotional

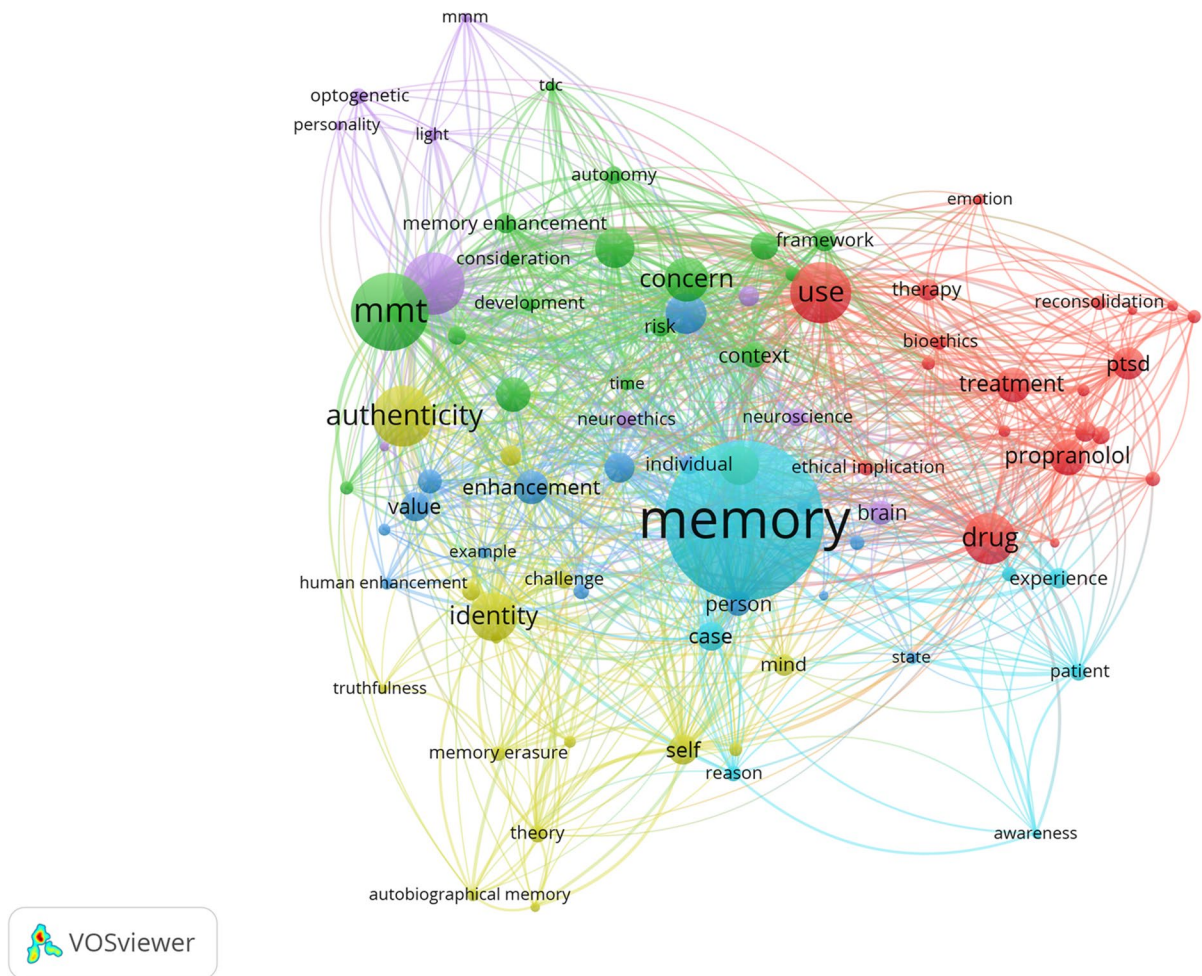


Fig. 4 Co-occurrence map of keywords in the field of MMT ethics

Table 3 Theme codes and reference quantities

Name	References
memory	2280
traumatic memories	88
emotional memories	45
painful memories	42
autobiographical memories	39
bad memories	35
memory consolidation	35
human memory	31
memory enhancement	30
declarative memory	28
memory modification	28
moral	506
moral responsibility	32
moral judgment	19
moral assessment	13
moral reaction	12
moral reasoning	12
moral obligation	11
moral values	9
morally permissible	9
moral behavior	8
moral duty	8
ethical	484
ethical issues	49
ethical concerns	43
ethical implications	37
ethical questions	21
ethical considerations	19
ethically permissible	13
ethical debate	12
ethical discussion	11
ethical problems	11
biomedical ethics	9

attitudes by paragraph, it can be determined that there are 2065 (15.2 per cent), 3453 (2.3 per cent), 3386 (24.9 per cent), and 4718 (34.6 per cent) reference paragraphs respectively presenting highly positive, moderately positive, moderately negative, and highly negative attitudes towards MMT (see Fig. 5).

Moreover, many scholars (Bush 2006; Cabrera and Elger 2016; Voarino et al. 2017; DePergola II 2018a; Glannon 2022) have mentioned that the research and clinical application of memory modification technology should adhere to the four bioethical

principles proposed by Beauchamp and Childress (2019), namely autonomy, beneficence, nonmaleficence, and justice. Among them, fifty-eight pieces of research (43.6 per cent) examined whether MMT would weaken human autonomy, specifically investigating whether MMT aligns with the ethical principle of respecting autonomy. Ninety-six pieces of research (72.2 per cent) examined whether MMT would promote health improvement, aligning with the principle of beneficence. Eighty-six pieces of research (64.7 per cent) mentioned concerns about the potential risks of MMT, aligning with the nonmaleficence principle. Sixty-five pieces of research (48.9 per cent) examined whether MMT would contribute to the widening of injustice, aligning with the principle of justice. In Western biomedical ethics research, this analytical framework is one of the most fundamental paradigms, but there are also some issues that cannot be fully encompassed within this framework. For instance, ninety-six pieces of research (72.2 per cent) explored issues such as whether MMT would lead to the loss of authentic self or the weakening of personal identity (Erler 2011; Zawadzki and Adamczyk 2021a; Leuenberger 2022).

Ethical Stances Analysis

In the previous section, I provided an outline of the ethical topics covered in the 133 literature sources examined in this study. On this groundwork, I will now explore how various scholars perceive the ethical problems of MMT across six themes in bioethics, including (1) self, identity, and authenticity; (2) autonomy and informed consent; (3) welfare and happiness; (4) safety and risks; (5) responsibility and obligation; (6) social and legal justice. Here it is necessary to make a clarification: these six themes are not completely disconnected; in fact, they overlap and relate to each other in certain topics, revealing the complex facets of MMT ethical issues.

Self, Identity, and Authenticity

Memory is seen as an important part of self. According to John Locke (1979), self is determined by memory. Jonathan Foster (2008) asserts distinctly that “you are your memory.” These words seem to provide us with an intuition that changing autobiographical

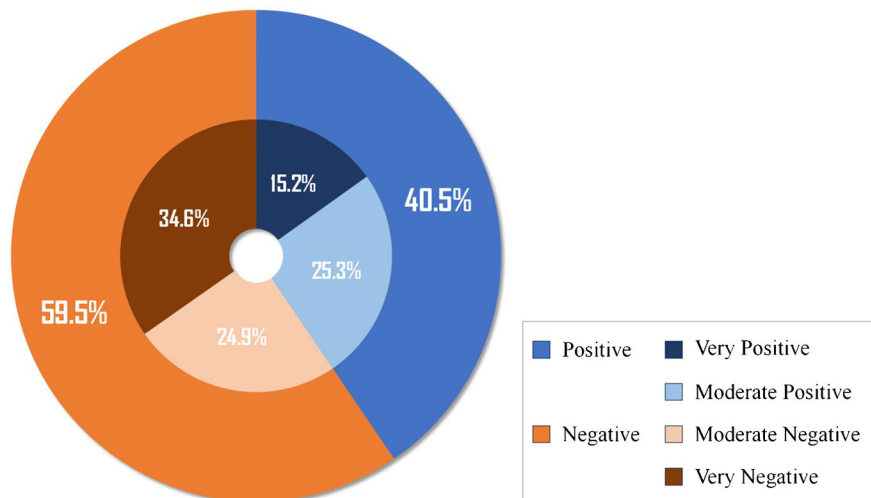


Fig. 5 The ratio of reference paragraph quantities of automatically coded emotions

memory through MMT may have implications for self, identity, and authenticity (Liao 2017). So, when facing the intervention of MMT on memory, the questions that ethicists need to consider are: Will MMT alter the self? Is the self of pre-MMT and post-MMT still the same person? And can the true self be preserved after MMT (Adamczyk and Zawadzki 2020; Dranseika, 2021)? These are the concerns regarding the self, identity, and authenticity in the application of MMT.

If we understand the self from the physiological perspective, where self-consciousness can be seen as a neural function of the brain, many forms of MMT will alter neural activity and thus have an impact on the self. For instance, Lavazza (2021) argues that autobiographical memory is “composed of multiple circuit activations,” so interventions on specific memory may affect other parts of brain, consequently impacting the self. If we acknowledge this, these changes in the brain and self may imply a disruption of our personal identity (Glannon 2019). However, some ethicists who do not believe that MMT will disrupt personal identity argue that the human brain is plastic, tolerant, and adaptive, so altering the brain does not necessarily mean that identity is irretrievably disrupted (Kolber 2006; Kostko 2016; Voarino et al. 2017; Kostick and Lázaro-Muñoz 2021). Additionally, those who take a more neutral stance believe that the physiological effects of MMT vary from individual to individual, and it cannot be conclusively

determined whether MMT will change personal identity (Henry et al. 2007; Lunstroth 2021).

If we approach this issue from a psychological perspective, there are two dimensions to consider in assessing the impact of MMT on the self: narrative and emotion. The narrative dimension implies that MMT intervenes in memory content, thus altering personal narrative, and since the self is based on narrative memory, MMT can induce changes in narrative self (Vukov 2017; Zawadzki 2022). The emotion dimension indicates that MMT intervenes in the emotional expression related to memory, thereby changing the self (Adamczyk and Zawadzki 2020). So, the key question is whether these influences on the narrative and emotional self undermine personal identity.

From the narrative dimension, those who argue that MMT undermines identity believe that the criterion for identity is narrative identity. That is, changes in the narrative memories of MMT users would disrupt their identity, thus rendering it morally unacceptable (Erler 2011; Racine and Affleck 2016). This loss of narrative identity seems to affect one’s understanding of oneself and alters responses to past events, thereby undermining authenticity (Vukov 2017). Another reason is that the self which is altered by narrative changes might alter past values, leading to a person based on false narratives with corresponding false values (Zawadzki 2022). With some assumptions about the inherent value of authenticity, the disruption of authenticity brought by MMT is morally

unacceptable. From the emotional dimension, critics of MMT also point out that drugs like propranolol can suppress the expression of negative emotions (Hurley 2010a, b), thus suppressing the self (Kabasenche 2007), leading to the rupture of personal identity (Pugh and Maslen 2017). For instance, according to Parens (2010), inability to express genuine emotions results in living in an alienated world, thereby losing the authentic self.

However, supporters of MMT have countered these arguments. Firstly, for some individuals, the “sting” of painful memories can be immense, potentially leading to a loss of identity and authenticity (Wasserman 2004). MMT interventions seem capable of restoring or even promoting identity and authenticity in this case (Gligorov 2016, 75–94; Hurley 2010a). Secondly, different approaches to MMT intervention may have varying impacts on identity and authenticity. A gentle, gradual intervention can still maintain narrative and emotional identity (Gray 2021; Lavazza 2018). Moreover, drugs like propranolol have decades of experience in treating cardiovascular diseases, with no evidence suggesting impacts on the identity or authenticity of these patients (Lázaro-Muñoz and Diaz-Mataix 2016).

Thus, another question arises: what are the reasons and core factors behind the divergence of opinions between the two camps? On one hand, unclear concepts and confusing definitions play a role. The divergence between the two camps may stem from different understandings of the self, leading to different interpretations of identity and authenticity. For instance, there are two distinct understandings of person: essentialism and existentialism (Iftode et al. 2022; Leuenberger 2022; Mihailov et al. 2021). Essentialism emphasizes self-discovery, while existentialism emphasizes self-creation (Pugh and Maslen 2017; Zawadzki 2022). If one acknowledges an essentialist self, the criterion for personal identity is rigid identity, maintaining a stable core; if one acknowledges an existentialist self, the criterion is extended identity, emerging in social interactions (Lavazza 2019). If personal perception of authenticity aligns with the rigid version, authenticity is something that needs to be conserved, and thus MMT may weaken authenticity; if one values the process of self-creation, authenticity takes on a “process”-oriented nature, and modifications by MMT may not disrupt authenticity (Leuenberger 2022).⁴ Faced with

the difficulty of ethical judgements due to confusion in current concepts, some scholars propose pluralist understanding of self, identity, and authenticity (Erler 2021; Tan and Lim 2020), using more flexible definitions to distinguish the different impacts of MMT on self, identity, and authenticity in different contexts, while rejecting rigid evaluative criteria (Gligorov 2016, 75–94; Lavazza 2016).

On the other hand, the values of self, identity, and authenticity are still not clear. Implicit in the aforementioned controversy is the assumption that maintaining “accurate” memories holds intrinsic value, and we should uphold this value (Rosenberg 2007; Phillips 2008; Erler 2011). However, the source of this value is ambiguous (Bublitz and Merkel 2009). Kroes and Liivoja (2019) argue this value stems from evolutionary adaptability, while Robillard and Illes (2016) believe it comes from the ability to navigate one’s position in the world, which is important for both oneself and others. Therefore, identity and authenticity are crucial criteria in assessing the ethical problems of MMT. However, Sandberg and Liao (2008) argue that some lies of memory in our lives are not absolutely terrible, and even if MMT disrupts identity and authenticity, it may not be ethically unacceptable because some other concerns may prioritize in some cases. Furthermore, autobiographical memory and life histories themselves contain many errors and biases (Erler 2022; Gligorov 2016, 75–94; Leuenberger 2022), which may make excessive demands on identity and authenticity inappropriate.

Autonomy and Informed Consent

Memory modification technology, as an intervention method, exerts control over an individual’s memory capacity and content. If autonomy is understood as self-rule or self-governance, then the intervention by MMT has the potential to increase personal autonomy in controlling and governing their own affairs because

⁴ It should be pointed out that although the internal logic of this dualist dichotomy is coherent, this framework seems problematic when we realize that the self itself may integrate many different forms in different contexts, and it may manifest certain forms of self more prominently. Appreciation is extended to Reviewer 2 for recommending clarification on this issue.

memory will function or fail to function according to the expectations of MMT users (Adamczyk and Zawadzki 2020). However, some scholars argue that MMT undermines human autonomy, making it morally unacceptable, and they provide three main reasons based on different understandings of the concept of “autonomy.”

The first argument suggests that autonomy refers to positive liberty, the ability to act in accordance with one’s desires. In this view, MMT is seen as a passive intervention that restricts the agency and initiative of the subject, preventing them from acting according to their own wishes (DePergola II 2018a; Vukov 2022). The second argument adopts the concept of negative freedom for autonomy, asserting autonomy without coercion. From this perspective, MMT is seen as a technology that promotes paternalism, suggesting that individuals may undergo memory modification under the coercion of others, thereby undermining their autonomy (Lavazza 2016, 2019; Voarino et al. 2017; Zawadzki and Adamczyk 2021b). Especially considering that individual adoption of MMT may be a form of societal pressure, leading users to develop illusions of autonomy, which is morally unacceptable (Fox et al. 2016). The third argument, taking a perspective of “relational autonomy,” posits that individual autonomy is established within social relationships. Since personal use of MMT can only alter one’s memory and not the memories of others, it is believed to cause individuals to relinquish their social relationships established in the past, thereby weakening their autonomy (Zawadzki 2022).

Moreover, the enhancement of autonomy for individuals using MMT may undermine the autonomy of others within the broader collective (Lavazza 2015). On the other hand, proponents of MMT argue that it offers individuals the freedom to selectively choose their memories, determining which ones to retain and which to discard (Mihailov et al. 2021). People can also choose for themselves the means by which to modify their memories (Moses 2019). This stance, which supports bio-liberalism and opposes bio-conservatism, contends that MMT promotes human autonomy, even endowing individuals with superagency (Wasserman 2004; Bostrom and Sandberg 2009; Farina and Lavazza 2022; Vukov 2017). Faced with such optimism, Kolber (2014) warns us that in the context of MMT, we can only advocate for limited freedom, with safety being the primary concern.

Furthermore, another central aspect of the autonomy issue revolves around the problem of informed consent in clinical context (Chandler et al. 2013). Some ethicists express concern that in the application of MMT, users may not fully comprehend the potential risks involved, thereby undermining their autonomy (Kolber 2006; Glannon 2007; Cabrera and Elger 2016; DePergola II 2018a; Erler 2021). For example, patients with PTSD may, due to their emotional and cognitive dependencies, make hasty decisions without gaining sufficient information about the risks of MMT, thereby diminishing patient autonomy and their right of self-determination (Bell 2008; Bublitz and Dresler 2015; Lavazza 2018; Glannon 2022). Additionally, Glannon (2014) provided a case where a patient, upon waking from general anesthesia in the operating room, overreacted to the bad news of biopsy, which shouldn’t have been known to the patient at this time. Consequently, the anesthesiologist administered propofol to the patient based on the preoperative consent regarding anesthesia, resulting in the patient forgetting the news upon awakening again. This also serves as a warning that we need to broaden the scope of patient informed consent to ensure that MMT does not encroach upon autonomy even in emergency situations (Tenenbaum and Reese 2007).

While in a military context, ethical dilemmas arise around whether soldiers can autonomously choose to forget traumatic memories and whether they can undergo compulsory memory enhancement, making it a contentious issue (Hurley 2010a, b; Donovan 2010; Liivoja and Kroes 2020). Experimental philosophy research shows that most people are in favour of allowing soldiers to have their memories modified (Newman et al. 2010) and believe individual soldiers should not bear the burden of suffering due to decisions made by politicians (Garren 2022). However, some ethicists still argue against granting soldiers the right to autonomously delete memories of war, as this involves a certain responsibility for collective memory or may lead to soldiers engaging in wrongful uninhibited behaviours (Kolber 2011).

Welfare and Happiness

In the report from the President’s Council on Bioethics (2003), it is mentioned that having better memory could lead to greater happiness, albeit in a form of

self-deception. In other words, on one hand, we strive for the authenticity of our memories and do not want ourselves to be deceived; on the other hand, in the pursuit of happiness and welfare, we tend to modify and change our memories, especially for those bad ones (Bublitz and Dresler 2015). For example, for individuals suffering from PTSD and burdened by traumatic memories, MMT appears to offer a potential way to alleviate suffering (Dekkers and Rikkert 2007; Evers 2007; Henry et al. 2007). The key issue is whether, from a utilitarian perspective, MMT really enhances the welfare of patients (Liao and Sandberg 2008).

Supporters of MMT argue that MMT can make people's minds happier and enhance cognitive function, thereby promoting health and leading to a more meaningful life; therefore, MMT can be morally acceptable (Bush 2006; Phillips 2008; Bostrom and Sandberg 2009; Muriithi 2014; Alonso Fernández 2020; Glannon 2022; Lavazza 2016). Meanwhile, opponents of MMT have engaged in deeper reflection on the argument of "false" or "superficial" happiness brought by MMT (President's Council on Bioethics 2003; Bublitz and Dresler 2015). Erler (2011) devises Elizabeth's case as a thought experiment to examine ethical issues related to MMT. In this case, Elizabeth, who had suffered from school violence throughout high school, carried this traumatic memory into adulthood. Seeking relief from the emotional distress caused by these memories, she opted for the assistance of MMT. It seems that MMT has improved Elizabeth's overall welfare (Lavazza 2019). However, Erler (2011) points out that this kind of improvement in well-being may not necessarily mean that the use of MMT can be morally accepted. In the case of Elizabeth, she could have the opportunity for personal growth from this traumatic memory, giving it a more positive meaning and thus achieving greater welfare (Erler 2011; Vedder and Klaming 2010; Warnick 2007). Zawadzki and Adamczyk (2021b), on the other hand, evaluated welfare in this context from the perspective of agency, arguing that the negative memories experienced by an individual may be so painful that she loses agency, and that MMT may be the only way she can regain agency and control over her life, which could enhance well-being as a result of a combination of individual values and societal ideology.

This case suggests that those memories that may appear bad also contain the value of happiness. If we

accept this point, then MMT may "undermine one's ability to seek, identify, and act on the good" in the long run (DePergola II 2018b). In other words, in the analysis of the welfare effects of MMT, it is necessary to comprehensively consider the user's short-term and long-term welfare, analysing changes in welfare over time and in response to environmental changes (Erden and Brey 2023).

The contemplation brought to us here is whether we should prioritize welfare and happiness when examining the ethical issues of MMT. Some scholars point out that considerations for enhancing health and welfare should take precedence (Liao and Sandberg 2008; Fletcher et al. 2010; Tan and Lim 2020). However, Miah (2009) and Glannon (2007) disagree with this welfare-first moral stance, emphasizing the need for a more comprehensive and cautious consideration of consequences of using MMT.

Another discussion related to welfare and happiness is the distinction between treatment and enhancement in the clinical application of MMT (Dekkers and Rikkert 2007; Elsey and Kindt 2016). The question of whether the state of a certain memory should be treated as a disease or as a normal response to an event and whether MMT is aimed at treating diseases or enhancing human capabilities is crucial for ethical judgements regarding MMT. Lavazza (2016) and Liao and Sandberg (2008) also point out that we need to balance subjective and social welfare with other moral values, and thus, there is no fixed standard for assessing the moral acceptability of MMT from the perspective of welfare.

Safety and Risks

As an emerging technology, MMT has drawn significant attention to its potential safety hazards and side effects. Bostrom and Sandberg (2009) have pointed out that, "risks accompany any intervention." Based on the principle of non-maleficence in bioethics, it is necessary to thoroughly examine the potential harm that MMT may cause to users, in other words, the potential risks. Therefore, it is essential to consider whether the benefits of MMT outweigh the known and potential risks for a comprehensive evaluation of its ethical acceptability (Hurley 2010a, b; Glannon 2014; Fox et al. 2016; Kroes and Liivoja 2019; Adamczyk and Zawadzki 2020).

Here we discuss MMT in both pharmaceutical and invasive forms separately. Memory-modifying drugs, according to some bioethicists (Carter and Hall 2007; Farina and Lavazza 2022), are highly safe and effective, suggesting that the President's Council on Bioethics may have exaggerated their harms. Pharmaceutical MMT, such as the use of propranolol, is considered to be able to significantly save time and money compared to traditional psychotherapy, and its effectiveness is more certain (Sententia 2006; Carter and Hall 2007; DePergola II 2018c). However, opponents of memory modification drugs warn us about the associated risks. Firstly, although these drugs demonstrate memory-modifying effects in model organisms, scientists find it challenging to ascertain whether they have adverse effects on animal autobiographical memory. Therefore, the transition from drug experiments to clinical applications poses safety concerns (Rose 2002; Tan and Lim 2020; Tang 2023). Secondly, it has been noted that many memory modification drugs may be addictive, posing a risk factor for users (Fox et al. 2016). Moreover, in treating PTSD, both pre-exposure prevention and post-exposure treatment approaches exist, but the comparison of their long-term potential risks is unclear, with insufficient evidence of their effectiveness, necessitating a cautious approach to memory modification drugs (Kroes and Liivoja 2019; Lázaro-Muñoz and Diaz-Mataix 2016). However, supporters of pharmaceutical MMT argue that hesitation due to potential risks may delay treatment for patients who urgently need it (Kolber 2011). Although drugs like propranolol may also impose cardiovascular burdens, pharmaceutical MMT appear to be safer overall compared to other antidepressant and anti-anxiety medications used in treating PTSD (Hall and Carter 2007; Hurley 2007; Liivoja and Kroes 2020).

Regarding memory modification through invasive neural technologies, some additional ethical challenges concerning safety and risks are encountered (Adamczyk and Zawadzki 2020, 2021; Zawadzki and Adamczyk 2021a; Gilbert et al. 2021). For instance, optogenetics as a genetic therapy requires the injection of a virus that modifies genes into host cells to integrate DNA molecules and achieve gene modification. However, this therapy has seen several cases of severe adverse reactions, raising concerns about its application (Adamczyk and Zawadzki 2020). But according to Gilbert et al. (2021), the risks associated

with optogenetic MMT are not uncontrollable at present. In the face of this counterargument, Adamczyk and Zawadzki (2021) agree that there are now many technological ways to circumvent as much as possible the potential risks of invasive neurotechnologies such as optogenetics, but other ethical issues associated with MMT remain complex. Additionally, considering the potential risks, a preference may be given to reversible rather than irreversible, and non-invasive rather than invasive, MMT to reduce associated risks (Cabrera and Elger 2016). For bioethicists, it is essential to find a balance between benefits and risks and to examine the equilibrium point between long-term and short-term benefits in order to ensure the ethical acceptability of MMT (Burbiel 2015). And for neuroscientists, we need more empirical evidence to verify whether (or which kind of) MMT is safe (Cabrera and Elger 2016).

Responsibility and Obligation

In the ethical discussions surrounding MMT, the issues of responsibility have also garnered much attention. Some point out that there exists a subtle tension between individual welfare and social responsibility (Kolber 2007; Tenenbaum and Reese 2007). Concerning MMT, ethicists are primarily focused on two responsibility issues: one pertains to the responsibility concerning memory itself, exploring whether memory modification eliminates people's responsibility for their memories; the other focuses on the additional responsibilities triggered by memory, investigating whether memory modification eradicates people's other moral responsibilities. If the answers to these two questions are affirmative, then we should not commit to the ethical acceptability of MMT.

Regarding the first responsibility issue, Kabasenché (2021) argues that memory is not just a cognitive ability or perceptual content; in many cases, due to the intertwining of individual life events with those of others, memory itself carries intrinsic responsibilities. As highlighted by the President's Council on Bioethics (2003, 232), "without memory, both our own and that of others, the notion of moral responsibility would largely unravel." For example, do witnesses have the responsibility not to forget the crime scene? Do soldiers involved in war have the responsibility not to forget their experiences on the battlefield? Do witnesses of holocaust and other

collective traumatized events have the responsibility not to forget those tragic memories (Hurley 2010a, b; Kolber 2006, 2007; Liao and Sandberg 2008; Cabrera and Elger 2016; Tillman 2016)? The social attributes carried by individual memories make memory modification a threat to moral responsibility. In other words, we seem to have an obligation not to modify memories, both for ourselves and for others (Bublitz and Dresler 2015; Bostrom and Sandberg 2009).

As for the second responsibility issue, which can be the possible sabotage of other moral responsibilities due to memory modification, ethicists opposing MMT argue that memory modification may diminish people's capacity to be responsible for their actions (Bublitz and Dresler 2015; DePergola II 2018c; Glannon 2019). One reason is that memory modification can influence a person's moral emotions and values, resulting in negative impacts on their moral behaviours (Cabrera 2015). For example, memory deletion can make individuals emotionally indifferent to negative events, eliminating feelings of regret and guilt, thereby affecting their moral behaviour and turning them into morally irresponsible actors, which can be deemed unacceptable (Cabrera and Elger 2016; Pugh and Maslen 2017). Another reason is that individuals who have lost certain memory fragments cannot be held accountable for the actions of their previous selves, and this discontinuity in self may lead to an inability to assume moral responsibility (DePergola II 2018b; Tan and Lim 2020). However, some scholars disagree with this viewpoint, arguing that the relationship between memory and moral responsibility is not as straightforward. Especially in cases of memory attenuation, considering the harm caused by painful memories, memory responsibility is not a decisive value factor (Rosenberg 2007; Liivoja and Kroes 2020).

Social and Legal Justice

In bioethical studies, justice issues are often understood as problems related to resource allocation, often referred to as “distributive justice” (Beauchamp and Childress 2019; Smith 2010; Baumrucker et al. 2012). Bioethicists need to address questions about who should receive the resources and who should bear the costs (Bush 2006). Specifically, if MMT exacerbates social injustice, then MMT becomes a

morally unacceptable technology (Sabini and Monterosso 2005; Hui and Fisher 2015; Lavazza 2019).

Regarding the impact of MMT on justice, the President's Council on Bioethics (2003, 232) holds a negative view and states, “Without truthful memory, there could be no justice or even the possibility of justice.” Specifically, some scholars point out that if MMT is economically accessible to the majority, its use could enhance social fairness and equality (Fox et al. 2016; Voarino et al. 2017). However, on the contrary, invasive neural technologies such as deep brain stimulation are currently expensive in clinical practice, which might widen the wealth gap and create neural inequality among socioeconomically unequal groups (Fox et al. 2016; DePergola II 2018a; Edwards 2020). Additionally, in cases of memory enhancement, individuals who are more likely to afford MMT may possess superior capabilities, potentially exacerbating societal inequality (Erden and Brey 2023). Moreover, using memory modification drugs may involve the issue of overmedicalization and is associated with the interests of various parties such as pharmaceutical companies and doctors (Henry et al. 2007; Sade 2007). On the other hand, Henry et al. (2007) argue that memory modification drugs like propranolol are much cheaper compared to other antidepressants and cognitive behavioural therapies, and their use in PTSD treatment does not necessarily contribute to widening social injustice. Meanwhile, Henry et al. (2007) introduced further concerns about social justice, suggesting that pharmaceutical companies “overmedicalize” memory-related problems for profit, implying the injustice in exploiting human suffering. Sade (2007), on the other hand, is displeased with such accusations, asserting that profit-seeking is not inherently unjust, especially considering the significant contributions of the pharmaceutical industry to healthcare.

In legal studies, justice is more commonly understood as respecting laws from the perspectives of “procedural” justice (Aksoy and Elmali 2002; Morreim 1995). On the one hand, using MMT to erase memories would simultaneously eliminate the user's sense of remorse and guilt. Whether this is reasonable in a judicial environment that seeks to enforce punishment is a rather complex issue (Snead 2011; Kreitmair 2016; Pugh and Maslen 2017). On the other hand, with the widespread use of MMT, people's social identities will undergo changes, and

their understanding of morality will be rewritten. The standards for ethical choices and value judgments may also change as memories are modified. For example, individuals, such as rape victims, who have altered memories and eliminated emotional responses, may no longer resist injustice (Tillman 2016; Vukov 2017; Adamczyk and Zawadzki 2020; McGorry 2017; Cadwallader 2016). Therefore, it is also necessary to implement social policies and legal regulations regarding MMT to ensure social fairness and justice (Henry et al. 2007).

In addition, Vedder and Klaming (2010) discuss the legal justice problem of memory enhancement in the court. Should it be acceptable for eyewitnesses at crime scenes to undergo memory enhancement, especially considering their need to testify in court and ensure the reliability of their testimony? Supporters argue that eyewitness memories are often unreliable, thus eyewitnesses should enhance their memories to ensure that the rights of others are not infringed due to the possible errors and biases in their memories, which can promote the common good (Vedder and Klaming 2010; Hauskeller 2010; Pierce 2010). Conversely, opponents point out that memory enhancement may come with the influence of other memories, which may not serve as good evidence (Lucke and Hall 2010; Nunn 2010). Moreover, it is doubtful whether eyewitnesses with enhanced memories are more trustworthy than those with normal memories (Lindsay 2010), and this form of MMT could be seen as a violation of personal values by the judiciary (Capps and Carter 2010; Chang and Buccafurni 2010; Tsu 2010).

Discussion

In summary, the academic community has extensively discussed the ethical issues in memory modification technology, covering various modification methods and different moral stances. However, there are still some flaws in current discussions on the ethical issues of MMT. Firstly, many ethicists use some principles and concepts in moral philosophy that are not clearly defined, leading to inconsistencies in addressing the same issues. Secondly, the ethical analysis of MMT primarily focuses on self and welfare issues, and there are relatively fewer discussions on the issues of responsibility and justice. Thirdly, the discussions

are concentrated on forms of modification involving optogenetic and pharmacological interventions, with insufficient attention given to emerging technologies such as deep brain stimulation and virtual reality that intervene in memory. Lastly, since MMT has not yet been widely and extensively deployed, the discussion resources for many scholars consist of thought experiments or conceptual analyses, creating a certain distance from moral practices in the biomedical practices.

Based on the current research status, I propose three suggestions to promote future ethical discussions and moral applications of MMT.

Firstly, we call for deeper reflections on the philosophical foundations of MMT. This involves ontological questions about what memory really is, epistemological questions about how we know and believe our memory, and axiological questions about what the value of memory is. As we gain insights into the molecular neural mechanisms of memory from cognitive neuroscience and become aware of the prevalence of false memories in our life experiences through experimental psychology, the philosophical foundations of memory pose even more intricate puzzles. Clarifying these fundamental issues is crucial for gaining sound ethical perspectives on MMT. If memory is merely the connections between neurons, then how should we understand the value of memory, and how should we consider the ethical consequences of modifying memory? If memory is more than just neural structures, then what type of memory does MMT modify, and what value does this kind of memory have? This also suggests that in the process of researching MMT from the perspective of applied ethics, we should undergo deeper reflection on the fundamental philosophical problems of memory from the perspective of philosophy of cognitive science and philosophy of mind.

Secondly, we call for a more precise clarification of ethical terms related to MMT. For example, which version of personal identity aligns better with our ethical scrutiny of MMT: narrative versions or numerical versions, self-discovery versions or self-creation versions? Which version of autonomy—active, passive, or relational—better characterizes our ethical requirements regarding MMT? We can acknowledge that the controversies surrounding these ethical terms can further explore the potential ethical impacts of MMT. However, we should also be cautious of the

misuse and compatibility of certain terms in our arguments. For instance, does the alternative common expression for memory modification, “memory manipulation,” carry certain value presuppositions? For technological neutralists, “manipulation” is simply understood as a process of “entirely or partially resetting the beliefs, desires, and other mental states of an agent,” constituting a form of intervention and influence (Noggle 2022). While in past ethical discussions, especially in fields like clinical ethics and artificial intelligence ethics, manipulation has often been considered morally unacceptable in presuppositions because of its close relationship with paternalism and coercion (Annoni and Miller 2016; Cohen 2018). So, the question that is worth further reflection is whether the “manipulation” in MMT is merely value-neutral “influence” or value-laden forms of “control”? Or should we explore alternative expressions such as “memory modulation,” “memory intervention,” or “memory adjustment”? Regardless, we should exercise greater caution in the use of terminology to avoid veering into off-topic discussions and biased discourse.⁵

Thirdly, we call for a more comprehensive empirical examination of the applications of MMT. Currently, human lifestyles and life experiences and even human nature are undergoing continuous reshaping due to the current rapid development of neuroscience and technology. Different forms of memory modification may have varying impacts on the neural structures associated with memory and internal phenomenological consciousness. We cannot simply assume that evaluating the outcomes of memory modification can conclusively determine the moral consequences of MMT. Instead, we need to conduct more detailed examinations involving technical nuances to provide a more comprehensive ethical reflection on social consequences (Lavazza 2015). Additionally, we should anticipate potential ways in which emerging neuro-technologies could intervene in memory. As Cabrera and Elger (2016) have pointed out, ethical reflection should precede the application of technology.

For techno-optimists, humanity has processed a novel kind of liberty: “neurorights” (Bublitz 2022; Ienca 2021). However, for techno-pessimists,

neural technologies bring about alienation, risks, and upheaval. It is foreseeable that in the future, humans will live in a societal landscape where memory is not inviolable. How should we embrace this “brave new world” of memory? Answering this question requires the collective involvement of neuroscientists, engineers, ethicists, and policymakers to better outline the future landscape of the human neural realm.

Finally, this study may have the following limitations. Firstly, in terms of literature retrieval, due to the intricate and diverse discussions in philosophy, ethics, and cognitive science related to memory, a small amount of research addressing ethical issues in MMT may have been overlooked. However, this study has minimized the risk of omitting important literature by employing systematic retrieval and expert recommendations. Therefore, it can be argued that this research has focused relatively comprehensively on the core literature of ethical issues related to MMT.⁶ Additionally, this research did not assess the merits and drawbacks of arguments by different scholars, a task that might require further research conducting a systematic study of different scholars’ arguments under various ethical themes.

Conclusion

This study is the first scoping review of the ethical issues related to memory modification technology within the scope of relevant literature. This article focuses on the ethical issues within six categories of concern in the field of memory modification technology. It presents the current academic positions, arguments, and controversies surrounding these issues. This prompts further reflection on the fundamental nature of memory and the relationship between humans and technology. As technology increasingly delves into the realm of our intrinsic experiential phenomena, there is a need for ongoing refinement of corresponding regulatory and governance systems to ensure that memory modification technology genuinely benefits humanity in a moral neuro-future.

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